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RESEARCH-ARTICLE

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Wuhan University of Technology

Published: 16 March 2024

[Citation in BibTeX format](#)

ICIAI 2024: 2024 the 8th International
Conference on Innovation in Artificial
Intelligence
March 16 - 18, 2024
Tokyo, Japan

Detection of greenwashing in ESG reports of Chinese listed companies based on Word2vec and TF-IDF

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ABSTRACT

With the growing emphasis on ESG (Environmental, Social, and Governance) issues, the mandatory disclosure of ESG reports is on the horizon. However, due to the lack of a regulatory framework and a unified international ESG evaluation, the phenomenon of greenwashing in corporate ESG reporting is prevalent. We collected social responsibility reports and actual ESG performance data from A-share companies from 2011 to 2021 and innovatively employed text mining techniques to quantitatively investigate the extent of greenwashing in ESG reports. Our study initially utilized the Word2Vec method, combined with Skip-gram and Continuous Bag of Words models to train word vectors, and built an ESG lexicon using seed words. ESG reports are subsequently segmented based on a defined sentence splitting function and TF-IDF algorithm is employed to extract keywords. By matching the keywords with the ESG lexicon, we precisely extracted the annual ESG discourse for each company and conducted sentiment analysis to derive a greenwashing score. Heterogeneity analysis reveals that firm ownership has no significant impact on the level of greenwashing, yet the industry and region in which the enterprise operates considerably influence the greenwashing level. This study holds implications for enhancing the quality of ESG reporting and optimizing investment decisions.

CCS CONCEPTS

• : Applied computing; • Enterprise computing; • Enterprise ontologies, taxonomies and vocabularies;

KEYWORDS

ESG, Greenwashing detection, Text mining, Word2vec, TF-IDF algorithm

ACM Reference Format:

Yan Chen and Ding Ma*. 2024. Detection of greenwashing in ESG reports of Chinese listed companies based on Word2vec and TF-IDF. In *2024 the 8th International Conference on Innovation in Artificial Intelligence (ICIAI 2024)*, March 16–18, 2024, Tokyo, Japan. ACM, New York, NY, USA, 6 pages. <https://doi.org/10.1145/3655497.3655513>

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ICIAI 2024, March 16–18, 2024, Tokyo, Japan

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ACM ISBN 979-8-4007-0930-2/24/03

<https://doi.org/10.1145/3655497.3655513>

1 INTRODUCTION

Since the beginning of the 21st century, Environmental, Social, and Governance (ESG) issues have risen to prominence as a crucial concern for both businesses and investors. Greenwashing behavior in ESG report refers to a behavior that is inconsistent in words and deeds in order to obtain the dividends of green development and promise some commitments to protect the environment or publish some news that is beneficial to oneself [1]. We leverage text mining technology to scrutinize extensive datasets of textual information from companies. Through this analysis, we compute a precise greenwashing score for each company, facilitating the identification of greenwashing behaviors. This approach aids investors and regulators in achieving a more accurate assessment of the authentic sustainability practices within enterprises. Ultimately, our efforts contribute to upholding the stability and fostering the healthy development of financial markets.

Text mining is an automated technology employed to extract valuable information and patterns from extensive textual data [2]. As early as the 1950s, H.P. Luhn introduced the concept of utilizing word frequency statistics for automated classification. In the 1960s, Gerald Salton and his colleagues introduced the vector space model and TF-IDF weighting, which formed the cornerstone of text retrieval. In the 2010s, the field of natural language processing witnessed the emergence of deep learning technology, giving rise to various models like Word2vec, BERT, and GPT. These models have gained widespread utilization in text mining, substantially enhancing the performance of text processing.

Text mining technology has found application in numerous domains. Liu et al. leveraged cutting-edge concepts and technologies in deep learning, including self-supervised learning and multi-task learning, to introduce an open-source financial pre-training language model named F-BERT, built upon the BERT model architecture. This model finds application in diverse downstream financial tasks [3]. Meng et al. employed ARMA-GARCH family models for the characterization of network sentiment and individual stock return series [4]. Zhao et al. introduced the Bayesian structure learning algorithm HDBNSL for on-board equipment, enhancing fault diagnosis accuracy within the high-speed railway signal system [5].

In the context of text mining models, Word2Vec stands out as a sophisticated approach. Word2Vec, an NLP tool introduced by Google researcher Tomas Mikolov and his colleagues in 2013. This tool is adept at capturing semantic information effectively and reducing dimensions. Word2Vec is well-suited for processing large-scale data, exhibits great versatility, and excels in various natural language processing tasks. Jiang et al. employed Word2Vec to autonomously acquire semantic information, established a recurrent convolutional neural network, and extracted profound semantic text features using its end-to-end network structure [6]. Utilizing

Word2Vec technology, An et al. extracted microblog topics for each stage of the event life cycle and conducted a collaborative analysis of microblog topics and emotions [7]. Evidently, Word2Vec is extensively employed in the field of NLP, serving as a potent tool for various natural language processing tasks.

The majority of existing literature has extensively examined the influencing factors and governance strategies pertaining to greenwashing behavior. However, there remains a notable gap in qualitative measurements specifically focused on assessing enterprise greenwashing practices. In our study, an ESG dictionary is constructed using Word2Vec, followed by text preprocessing via the TF-IDF algorithm. Subsequently, text mining is conducted on the social responsibility reports of listed companies to calculate their annual ESG greenwashing scores.

2 GREENWASHING SCORE GENERATION ARCHITECTURE

In order to identify the greenwashing behavior of listed companies, we established a greenwashing detection program based on text mining technology to calculate the greenwashing score of listed companies.

Before identifying greenwashing among publicly traded companies, it is imperative to establish an ESG lexicon to effectively extract pertinent ESG-related content from textual sources. Traditional methods for developing such a lexicon involve manual definition and translation from foreign dictionaries. However, these methods are susceptible to overlooking critical terms and concepts, and disparities between Chinese and foreign terminology can undermine word accuracy [8].

In this research, we have developed an ESG lexicon employing a multi-stage approach encompassing data preparation, pre-processing, feature extraction, representation, modeling, and interpretation. We have harnessed the Word2Vec technique to perform deep learning on the lexicon generated by word segmentation. Our aim has been to uncover ESG-related terms with significant inter-connections. Utilizing the Skip-gram and Continuous Bag of Words architectures, we have efficiently and swiftly trained word vectors, thereby acquiring a comprehensive ESG lexicon.

Next, we conduct text mining on the social responsibility report of listed companies. To begin, the reports undergo preprocessing through clause and word segmentation. Subsequently, the TF-IDF algorithm is employed to extract keywords from each sentence of the fundamental ESG corpus, which is then used to filter the ESG corpus of each enterprise and year. Finally, the Chinese sentiment lexicon developed by Jang et al. is introduced to evaluate the presence of Chinese sentiment words in the ESG corpus. By performing a series of calculations [9], the study obtains the greenwashing score of listed companies. The overall structure of the ESG greenwashing detection process for listed companies is depicted in Figure 1.

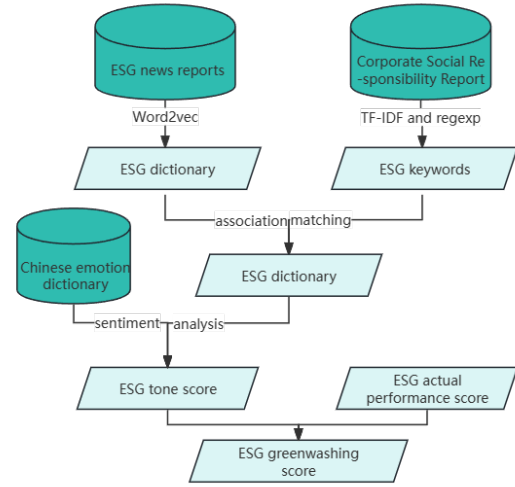


Figure 1: The overall architecture diagram of greenwashing detection

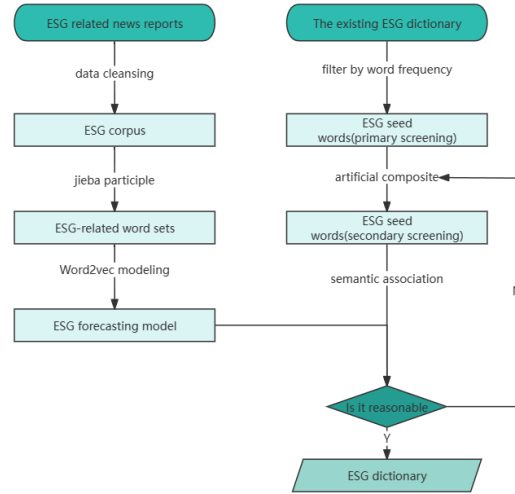


Figure 2: ESG dictionary construction flow chart

3 MODEL ESTABLISHMENT AND ALGORITHM IMPLEMENTATION

3.1 Word2Vec Builds ESG Dictionary

The text within the social responsibility reports of listed companies frequently includes extraneous information. To mitigate the impact of this data and enhance the precision of the greenwashing score, this study seeks to precisely identify ESG-related segments within the social responsibility reports of listed companies using an ESG dictionary. The ESG dictionary construction flow chart is illustrated in Figure 2.

Firstly, we begin with data preparation to retrieve ESG-related news reports from the Sogou Lab Corpus. Subsequently, we conduct

data cleaning to eliminate extraneous content, including advertisements, tags, HTML, JavaScript, and other codes and annotations. We extract ESG-related textual content, encompassing content categories, titles, and body text using regular expressions. Following this, we perform text preprocessing, utilizing the Jieba library for word segmentation. To enhance the accuracy of word segmentation, we import stop words and update the word segmentation dictionary within the Jieba library.

Following corpus preprocessing, the next crucial consideration pertains to the transformation of segmented words into a format amenable to computer-based calculations. It is evident that, in order to perform any computational operations, it is imperative to convert the segmented Chinese words from textual form into numerical representations, a process known as word vectorization. In the Skip-gram model, each word within the text serves as the focal word (target word), and the surrounding words function as the context. Conversely, in the Continuous Bag of Words (CBOW) model, the context word assumes the role of the target word, and the target word becomes the context [10].

In our model, the input layer is directly represented by a Bag of Words (BOW), and the One-Hot Encoding representation of contextual words is employed as input to directly aggregate vectors, reducing their dimensionality. Hierarchical Softmax is utilized to restructure the output into a hierarchical tree format. At each branching point, the problem is reformulated as binary classification, and the One-Hot Encoding representation of the target word is generated. We employ Negative Sampling and Hierarchical Softmax in multiple iterations to train the neural network, leading to a substantial reduction in computational complexity [11]. During the training phase, the weights of the hidden layer within the network serve as the word embedding vectors we aim to acquire. Upon completion of training, these hidden layer weights can be extracted from the neural network, providing the word embedding vector for each word. These vectors encompass information pertaining to word semantics and contextual context [12].

We compiled existing dictionaries related to ESG, and conducted initial screening based on word frequency statistics. ESG seed words were selected through manual secondary screening by an expert team. Employing a trained word vector model, we identified words exhibiting a robust semantic relationship with the provided ESG seed words, thus constructing an ESG keyword lexicon.

3.2 TF-IDF algorithm to extract keywords

TF-IDF (Term Frequency-Inverse Document Frequency) is a widely employed algorithm for extracting keywords from text. This research leverages the TF-IDF algorithm to identify the most crucial words within the social responsibility reports of publicly listed companies with respect to their sentence content.

The process includes setting the clause function, using the 'jieba' tool to segment the social responsibility report according to the punctuation, and structuring the resultant data into a list in list format, as depicted in Eq.1), for subsequent importation into a pandas data frame.

$$text = [[w_{1,1}, w_{1,2}, \dots], [w_{2,1}, w_{2,2}, \dots], \dots, [w_{n,1}, w_{n,2}, \dots]] \quad (1)$$

In this study, we construct a dictionary and create a bag-of-words (BoW) sparse vector representation for the document. To calculate

the Term Frequency (TF) value, we employ Eq.2), which is based on the word's frequency within sentence $count_{i,r}$ relative to the total word count in that sentence.

$$TF_{i,r} = \frac{count_{i,r}}{\sum_{n=0}^r count_{i,r}} \quad (2)$$

Inverse Document Frequency (IDF) serves as an indicator for gauging the significance of a word within an entire document collection. A higher IDF value denotes that a word is less common but more informative. The calculation of the IDF value for a word is based on both the total number of sentences and the total number of sentences in which the word appears within the document, as demonstrated in Eq.3).

$$IDF_{i,r} = \lg \frac{|ST|}{|SW_i| + 1} \quad (3)$$

Subsequently, the TF-IDF value for each word is computed by multiplying the word frequency by the inverse document frequency. Words with elevated TF-IDF values are recurrent within sentences but are infrequent across the entire document [13] [14].

$$TF - IDF_{i,r} = TF_{i,r} \cdot IDF_{i,r} \quad (4)$$

For every sentence, the first five words with the highest TF-IDF values are selected as its keywords. This process is iteratively applied to each sentence to extract their respective keywords.

3.3 Greenwashing score calculation

To ensure the retention of text data closely aligned with sustainability and enhance data quality by reducing noise, this paper employs the keywords from each sentence and the established ESG dictionary for association matching. This approach allows for the precise extraction of the annual ESG corpus for each enterprise. Following this, we introduced the Chinese sentiment dictionary developed by Jiang et al. to identify sentiment words within the text. We measure the positive and negative emotional words within the ESG corpus using a cyclic traversal method.

Drawing from the research conducted by Yu et al. in 2020, we posit that greenwashing serves as a facade for companies attempting to portray sustainable performance through selectively disclosed ESG data, yet falling short in actual ESG performance [15]. We use the frequency of positive emotional words minus the frequency of negative emotional words to get the ESG greenwashing tone. Furthermore, we procure the annual ESG performance data for each company from the Bloomberg database. To ensure data comparability, we employ Z-Scores for data standardization. The combination of Eq.5) and Eq. 6) yields the mean ESG greenwash tone and actual ESG performance, along with their respective standard deviations [16].

$$ESG_{-tone-stand} = \frac{ESG_{-tone} - \overline{ESG_{-tone}}}{\sigma_{ESG_{-tone}}} \quad (5)$$

$$ESG_{-act-stand} = \frac{ESG_{-act} - \overline{ESG_{-act}}}{\sigma_{ESG_{-act}}} \quad (6)$$

In accordance with Eq.7), we employ the disparity between the ESG greenwash tone and actual ESG performance to assess a company's greenwashing level. This approach aligns with our definition of greenwashing and offers a resolution to certain issues within the

prevailing literature related to the 'exaggerated' measurement of greenwashing problems [17].

$$\text{Greenwash Score}_{i,t} = \text{ESG-tone-stand}_{i,t} - \text{ESG-act-stand}_{i,t} \quad (7)$$

4 STATISTICAL ANALYSIS

This study focused on China's A-share listed companies spanning the period from 2011 to 2021. To align with our research objectives, we excluded ST companies subjected to special treatment during the sample period. Additionally, we excluded financial institutions, including banks and insurance companies, to account for industry-specific considerations. Lastly, companies that did not disclose their ESG performance were also excluded. Utilizing the aforementioned text mining greenwashing detection program, this research acquired a dataset comprising 6,714 greenwashing scores for 929 listed companies over the 2011-2021 period.

4.1 Time-trend analysis

In order to better understand the changing trend of listed companies' greenwashing data, predict the future situation, and provide information for the decision-making of investors, financial institutions and government regulators. We calculate the average values of ESG-tone , ESG-act and Greenwash Score of the overall sample in different years by arithmetic average method, and draw the time trend analysis chart as shown in Figure 3.

From the chart, it can be seen that the ESG-tone , ESG-act and Greenwash Score of the sample companies remained basically unchanged before 2015, and the actual performance level of ESG increased significantly in 2015, and then showed an upward trend. The overall tone of ESG also shows an upward trend but is relatively slow. Meanwhile, the Greenwash Score, a measure of deceptive or insincere ESG efforts, witnessed a substantial decline beginning in 2015 and has since continued on this downward trajectory. This decline in greenwashing can be attributed, in part, to China's environmental regulatory policies. In 2015, China issued the 'Water Pollution Prevention Action Plan' and the 'Solid Waste Pollution Prevention Action Plan'. These policy initiatives demonstrate a strong commitment to environmental protection and sustainable practices. It can be seen that the national environmental regulation policy can greatly reduce the greenwashing level of listed companies. Listed companies are highly responsive to the national environmental policy and have achieved positive results in promoting the construction of ecological civilization [18].

4.2 Sample analysis

In order to deeply understand the ESG greenwashing performance of different types of companies, the regional and industrial differences of ESG greenwashing performance are accurately analyzed. We collected data from the Wind database on the nature of the sample company, the industry, and the region, and analyzed the characteristics of companies with different ESG greenwashing levels.

Initially, we employ SPSS to conduct a Kruskal-Wallis test, with the aim of examining the distribution of greenwashing scores across various types of companies. Our categorization includes state-owned companies, private companies, foreign-funded companies,

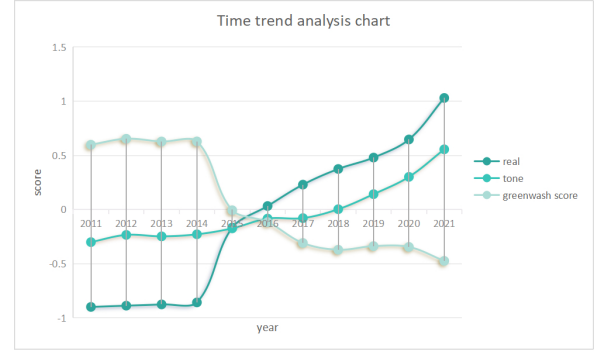


Figure 3: ESG greenwashing score time trend chart

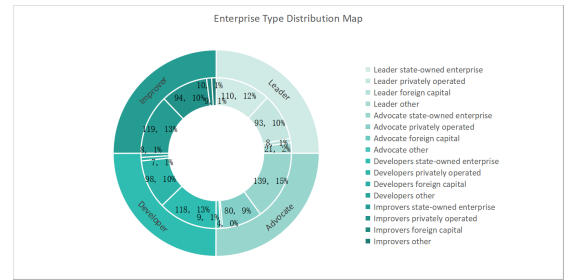


Figure 4: Enterprise type distribution map

and others. We rank the observations of each sample in an increasing order and calculate the rank of each data. R_i is the sum of the ranks of each group and n_i is the number of samples in each group. KW is calculated by Eq.8).

$$KW = \frac{12}{n(n+1)} \sum_{i=1}^K \frac{R_i^2}{n_i} - 3(n+1) \quad (8)$$

In cases where there are identical rank values within the common sample, it becomes imperative to implement an adjustment, as outlined in Eq.9). And the correction coefficient is as shown in the Eq.10).

$$Adj = 1 - \frac{\sum(\omega_j^3 - \omega_j)}{n^3 - n} \quad (9)$$

$$KW_{Adj} = \frac{KW}{Adj} \quad (10)$$

The Adj.Significant levels are presented in Table 1. Upon analysis, it becomes evident that, Significance and Adj.significance of state-owned companies and private companies, state-owned companies and foreign-funded companies are both 1. This implies that the distribution of greenwashing scores among state-owned companies and non-state-owned companies is identical. we employ a quartile-based method to categorize the sample companies into four distinct groups: ESG Leader, ESG Advocate, ESG Developer, and ESG Improver, based on their respective levels of greenwashing scores [19]. By analyzing the distribution of enterprise types in relation to the ratio of companies with varying ownership structures, we have constructed a visual representation. From Figure 4, we can see that the distribution of state-owned

enterprises, private enterprises and foreign-funded enterprises in the four ESG greenwashing levels is similar. This observation reinforces the conclusion that there is no significant difference in ESG greenwashing levels between state-owned and non-state-owned companies.

To examine the extent of ESG greenwashing across various industries, we have employed the national economic industry classification standard to categorize the sample companies. To ensure the relevance of our findings and mitigate the influence of outdated data, we have opted for recent data, providing insights into current trends and developmental trajectories. In Figure 5, we present the ESG greenwashing scores for companies in 17 different industries over the three-year period spanning from 2019 to 2021. Our analysis reveals that the K real estate industry and Q health and social work industry exhibit higher levels of ESG greenwashing. In contrast, the B mining industry, C manufacturing industry, and M scientific research and technology service industry, which have a higher likelihood of environmental pollution, demonstrate lower levels of ESG greenwashing.

We have conducted an analysis to elucidate potential factors contributing to the observed trends. It appears that the real estate industry is more prone to generating social challenges due to factors such as opaque corporate governance and market pressures favoring short-term profit maximization. These dynamics can incentivize ESG greenwashing practices within this sector. Conversely, the health and social work industry primarily centers on delivering social services, with lesser involvement in environmental and governance concerns. In contrast, industries such as mining, manufacturing, and scientific research and technology services are intrinsically linked to technological innovation and resource management. As a result, they are more inclined to adopt environmentally conscious and sustainable development practices. Furthermore, it is noteworthy that industries with a higher likelihood of causing pollution are subject to increased regulatory scrutiny. Consequently, companies operating within these sectors tend to have more comprehensive and accurate ESG information disclosures, enhancing transparency and veracity.

To assess the extent of ESG greenwashing across various regions, we computed the arithmetic average of ESG greenwashing scores spanning the years 2019 to 2021, and subsequently generated Figure 6. Between 2019 and 2021, Hainan, Heilongjiang, Inner Mongolia, and Liaoning exhibit comparatively high levels of ESG greenwashing. In contrast, regions such as Jiangxi, Henan, and Guizhou demonstrate relatively lower levels of ESG greenwashing, indicative of a positive trajectory towards sustainable practices.

Through our extended research, we have discerned that variations in economic structures and ESG regulatory policies among different regions serve as fundamental drivers of disparities in ESG greenwashing levels [20]. Notably, Heilongjiang and Liaoning stand out as regions characterized by an economic landscape predominantly comprised of heavy industries, heavily reliant on traditional and resource-intensive sectors. This dependence gives rise to heightened environmental shadow pollution and concomitantly leads to diminished actual ESG performance. In contrast, The low level of ESG supervision in Hainan and Inner Mongolia leads to the high tone of ESG. The economic structure of a region significantly influences the tangible ESG performance of businesses,

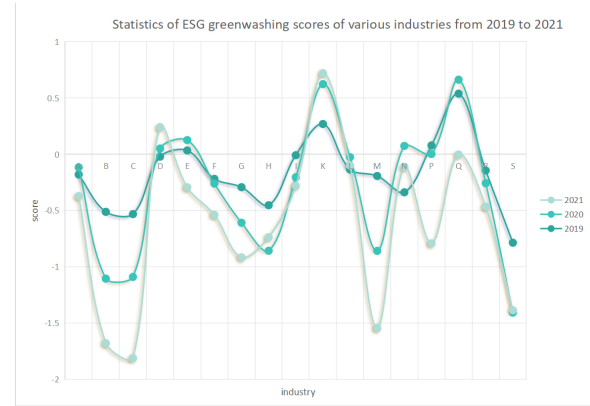


Figure 5: Statistics of ESG greenwashing scores of various industries from 2019 to 2021

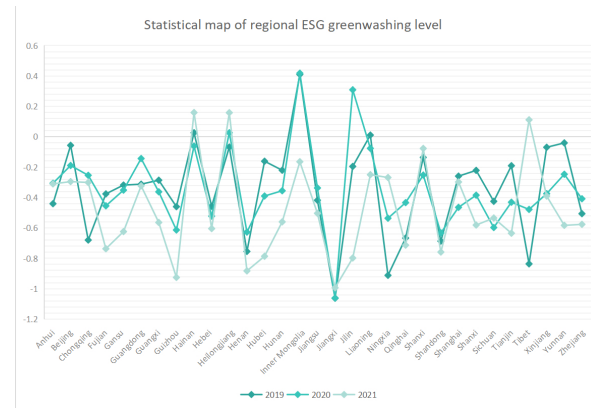


Figure 6: Statistical map of regional ESG greenwashing level

impacting resource dependence, the labor market, as well as innovation and technological dynamics. Meanwhile, ESG regulatory frameworks, encompassing ESG information disclosure systems and accountability measures within a given region, directly shape the ESG presentation of companies. Consequently, the interplay between regional economic structures and ESG regulatory policies jointly contributes to the level of ESG greenwashing within the region.

5 CONCLUSION

This study uses text mining technology to develop an ESG greenwashing detection tool. Our method first innovatively uses the Word2Vec method, combines Skip-gram and Continuous Bag of Words model for word vector training, and integrates seed words to build an ESG lexicon. Then, we segment the sentences of the ESG report, and select the top five words with the highest TF-IDF value in each sentence as keywords according to the calculated TF-IDF value of the words. Subsequently, we match these keywords with the ESG dictionary to achieve accurate extraction of the annual ESG-related corpus of the enterprise. Finally, the greenwashing score is calculated by sentiment analysis.

Table 1: Statistical map of regional ESG greenwashing level

Sample 1-Sample 2	Test tatistical	Standard error	Standard test	Significance	Adj.Significance
State-owned&Private foreign	-40.445	51.536	-0.785	0.433	1.000
State-owned&Investment	208.698	152.464	1.369	0.171	1.000
State-owned&Other	274.911	111.475	2.466	0.014	0.082
Private foreign&Investment	249.143	155.150	1.606	0.108	0.650
Private foreign&Other	315.355	115.122	2.739	0.006	0.037
Investment&Other	66.212	183.963	0.360	0.719	1.000

The results show that the state's environmental regulation policy significantly reduces the ESG greenwashing level of listed companies. In our subsequent sample analysis, no substantial disparity in ESG greenwashing levels emerged between state-owned enterprises and their non-state-owned counterparts. Simultaneously, entities operating within the technological innovation and resource management sectors exhibited a comparatively diminished proclivity for greenwashing. In contrast, those situated in heavily industrialized regions and regions characterized by lax ESG oversight demonstrated reduced levels of ESG greenwashing.

The ESG greenwashing detection program developed in this study holds significant promise for wide-ranging applications. Not only does it facilitate the quantification of environmental protection initiatives for businesses, but it also furnishes investors with reliable ESG metrics, thereby guiding more informed investment decisions. Concurrently, this empirical data empowers policymakers to devise more precise ESG regulatory frameworks. The pioneering tools for ESG greenwashing detection are poised to play a pivotal role in shaping the trajectory of sustainable development and enhancing China's ESG framework.

Future research can be further deepened from the following aspects: exploring the application of emerging technologies such as blockchain and artificial intelligence in greenwashing governance to improve the credibility, traceability and security of enterprise data and reduce corporate greenwashing behavior; investigate the impact of different economic incentive mechanisms on corporate greenwashing behavior; pay attention to the establishment of a unified ESG system, formulate incentive policies, and standardize laws and regulations to curb corporate greenwashing behavior.

ACKNOWLEDGMENTS

This research is supported by the National Natural Science Foundation of China (Grant No. 72004173).

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